CIIM4

QUADRATIC EQUATIONS

Introduction

(i) $P(x) = a_n x^n + \cdots + a_1 x + a_0$, $a_n \neq 0$ (polynomial of degree n)

(ii) $P(x) = 0$ is an **equation** ; true for all $x \Rightarrow$ **identity**

☆ (iii) Degree 2 : $\boxed{ax^2 + bx + c = 0}$, $a \neq 0$ (quadratic)

(iv) Root : $P(\alpha) = 0$; degree $n \Rightarrow n$ roots (real + complex)

(vi) Fundamental theorem : degree n has exactly n roots (counting multiplicity)

(v) Real coefficients : complex roots occur in conjugate pairs

Ex. $x^2 - 5x + 6 = 0 \Rightarrow (x-2)(x-3) = 0 \Rightarrow x = \boxed{2,\ 3}$

Key points in solving an equation

(i) Squaring both sides may add **extra roots** $\rightarrow$ put back and verify

(ii) Check the domain : denominator $\neq 0$, $\sqrt{\ } \geq 0$, $\log$ argument > 0

(iii) Common factor (with x) cancel mat karo $\rightarrow$ root lost

(iv) $|f(x)| = k \Rightarrow f(x) = \pm k$ $(k \geq 0)$; modulus ke liye cases banao

Ex. $\sqrt{x+2} = x$: $x^2 - x - 2 = 0$, $x = 2$ or -1 $\Rightarrow x = -1$ rejected $\Rightarrow$ $\boxed{2}$

Ex. $x + \dfrac{1}{x-1} = 1 + \dfrac{1}{x-1}$ $\Rightarrow x = 1$ but $x \neq 1$ $\Rightarrow$ no solution

Ex. $|x-2| = 3 \Rightarrow x - 2 = \pm 3 \Rightarrow \boxed{5,\ -1}$

Ex. $\sqrt{2x-1} = x - 2$: $x^2 - 6x + 5 = 0 \Rightarrow x = 1, 5$

$x = 1$ gives $1 = -1$ (rejected) $\Rightarrow$ $\boxed{5}$

Ex. $\dfrac{x^2 - 1}{x - 1} = 0 \Rightarrow x = \pm 1$ but $x \neq 1 \Rightarrow$ $\boxed{-1}$

Ex. $x^2 = x$: dividing by x loses a root $\Rightarrow x(x-1) = 0 \Rightarrow \boxed{0,\ 1}$

Remainder & Factor theorem

☆ (i) $P(x) = (x-a)\,Q(x) + R \;\Rightarrow\; \boxed{R = P(a)}$

(ii) $(x-a)$ is a factor $\iff P(a) = 0$

(iii) Divisor $(ax+b)$: $R = P\left(-\dfrac{b}{a}\right)$

(iv) Divisor $(x-a)(x-b)$: remainder $px+q$, use $P(a), P(b)$

Ex. $2x^3 - 3x^2 + x - 5$ divided by $(x-2)$: $R = P(2) = 16 - 12 + 2 - 5 = \boxed{1}$

Ex. $x^3 - 6x^2 + 11x - 6$: $P(1) = 0 \Rightarrow (x-1)(x-2)(x-3)$

Ex. $P(1) = 2$, $P(2) = 5$: remainder on $(x-1)(x-2)$ is $px+q$

$p + q = 2$, $2p + q = 5 \Rightarrow \boxed{3x - 1}$

Ex. x^{100} divided by $x^2 - 1$: $x^2 \equiv 1 \Rightarrow$ remainder $\boxed{1}$

Repeated roots

☆ (i) α is a repeated (double) root of $P \iff \boxed{P(\alpha) = 0 \text{ and } P'(\alpha) = 0}$

(ii) Root repeated m times $\iff P(\alpha) = P'(\alpha) = \cdots = P^{(m-1)}(\alpha) = 0$

(iii) Quadratic : equal roots $\iff D = b^2 - 4ac = 0$

Ex. $x^3 - 3x^2 + 4$: $P(2) = 0$, $P'(2) = 12 - 12 = 0$

$\Rightarrow (x-2)^2(x+1)$, repeated root $\boxed{2}$

Ex. $x^3 - 3x + 2 = (x-1)^2(x+2) \;\Rightarrow$ repeated root 1

Identity equations & inequalities

☆ (i) Degree $\le n$ polynomial with more than n roots $\Rightarrow$ identity (all coeffs 0)

(ii) $ax^2 + bx + c = 0$ for **3 distinct** values of $x \Rightarrow a = b = c = 0$

(iii) Identity $\Rightarrow$ compare coefficients of like powers

(iv) Lagrange : $\displaystyle\sum \frac{(x-b)(x-c)}{(a-b)(a-c)} = 1$ $(a, b, c$ distinct$)$

Ex. $ax^2 + bx + c = 0$ has roots $1, 2, 3 \Rightarrow \boxed{a = b = c = 0}$

Ex. $x^2 - 3x + 2 \equiv (x-1)(x+p)$ for all $x \Rightarrow$ compare : $p = \boxed{-2}$

Quadratic equation : $ax^2 + bx + c = 0$

☆ (i) Roots : $\boxed{x = \dfrac{-b \pm \sqrt{D}}{2a}}$, $D = b^2 - 4ac$

(ii) $D > 0$: real, distinct ; $D = 0$: real, equal ; $D < 0$: complex conjugates

(iii) a, b, c rational, D perfect square $\Rightarrow$ roots rational

(iv) a, b, c rational, $p + \sqrt{q}$ root $\Rightarrow$ $p - \sqrt{q}$ also a root

(v) a, b, c real, $p + iq$ root $\Rightarrow$ $p - iq$ also a root

(vi) $|\alpha - \beta| = \dfrac{\sqrt{D}}{|a|}$

Ex. $x^2 - 4x + 13 = 0$: $D = 16 - 52 = -36 \Rightarrow x = \dfrac{4 \pm 6i}{2} = \boxed{2 \pm 3i}$

Sum & product of roots

☆ (i) $\alpha + \beta = -\dfrac{b}{a}$, $\alpha\beta = \dfrac{c}{a}$

☆ (ii) Equation with roots α, β : $\boxed{x^2 - (\alpha + \beta)x + \alpha\beta = 0}$

(iii) $\alpha^2 + \beta^2 = S^2 - 2P$; $\alpha^3 + \beta^3 = S^3 - 3SP$; $(\alpha - \beta)^2 = S^2 - 4P$

(iv) $\alpha^4 + \beta^4 = (S^2 - 2P)^2 - 2P^2$; $\dfrac{1}{\alpha} + \dfrac{1}{\beta} = \dfrac{S}{P}$

(v) Recurrence : $\alpha^n + \beta^n = S(\alpha^{n-1} + \beta^{n-1}) - P(\alpha^{n-2} + \beta^{n-2})$

Ex. $x^2 - 5x + 3 = 0$: $\alpha^3 + \beta^3 = 125 - 45 = \boxed{80}$

Ex. Roots = squares of roots of $x^2 - 3x + 1$: $S' = 9 - 2 = 7$, $P' = 1$

$\Rightarrow$ new equation $\boxed{x^2 - 7x + 1 = 0}$

Ex. $x^2 - x - 1 = 0$: $\alpha^5 + \beta^5$: $2, 1, 3, 4, 7, \boxed{11}$

Sign of roots

(i) Both positive : $D \geq 0$, $S > 0$, $P > 0$

(ii) Both negative : $D \geq 0$, $S < 0$, $P > 0$

(iii) Opposite signs : $P < 0$ ($D > 0$ automatically)

(iv) $c = 0$: one root 0 ; $b = 0$: roots equal, opposite

(v) $a = c$: reciprocal roots

(vi) Roots in ratio $m:n$: $mnb^2 = (m+n)^2 ac$

Ex. $x^2 - kx + 4 = 0$ both roots positive : $D = k^2 - 16 \geq 0$, $S = k > 0$

$\Rightarrow$ $\boxed{k \geq 4}$

Ex. $x^2 - x - 6 = 0$: $P = -6 < 0 \Rightarrow$ opposite signs (roots $3, -2$)

Ex. $2x^2 - 7x + 3 = 0$: $\alpha^2 + \beta^2 = \frac{49}{4} - 3 = \boxed{\frac{37}{4}}$

Examples on quadratic equation

Ex. $x^2 - 6x + k = 0$: one root twice the other : roots $r, 2r$, $3r = 6$

$\Rightarrow k = 2r^2 = \boxed{8}$

Ex. $x^2 - 5x + k = 0$: roots differ by 3 $\Rightarrow 25 - 4k = 9 \Rightarrow k = \boxed{4}$

Ex. α, β roots of $x^2 - 3x + 1$: equation with roots $\alpha + \frac{1}{\beta}$, $\beta + \frac{1}{\alpha}$

$\alpha\beta = 1 \Rightarrow$ roots $2\alpha, 2\beta \Rightarrow$ $\boxed{x^2 - 6x + 4 = 0}$

Ex. $a + b + c = 0 \Rightarrow x = 1$ is a root ; $3x^2 - 5x + 2$: roots $1, \frac{2}{3}$

Ex. $x^2 + x + 1 = 0$ roots ω, ω^2 : $\omega^{100} + \omega^{200} = \omega + \omega^2 = \boxed{-1}$

Ex. $x^2 - mx + 6 = 0$ integer roots : $(1,6), (2,3), (-1,-6), (-2,-3)$

$\Rightarrow m = \boxed{\pm 5, \pm 7}$

Ex. $x + \frac{4}{x}$ for $x > 0$: AM $\geq$ GM $\Rightarrow$ minimum $\boxed{4}$ at $x = 2$

Quadratic inequalities

(i) $a > 0$, roots $\alpha < \beta$: $f(x) > 0 \Rightarrow x < \alpha$ or $x > \beta$

(ii) $a > 0$: $f(x) < 0 \Rightarrow \alpha < x < \beta$

(iii) $|x - a| < k \Rightarrow a - k < x < a + k$

(iv) $|x - a| > k \Rightarrow x < a - k$ or $x > a + k$

(v) Never multiply by an expression whose sign is unknown

Ex. $x^2 - 5x + 6 < 0 \Rightarrow (x-2)(x-3) < 0 \Rightarrow \boxed{2 < x < 3}$

Ex. $x^2 - 4x + 3 \geq 0 \Rightarrow \boxed{x \leq 1 \text{ or } x \geq 3}$

(vi) Rational inequality : one side 0, mark critical points, wavy curve

Ex. $\dfrac{(x-1)(x-3)}{x-2} \geq 0 \Rightarrow \boxed{[1,2) \cup [3,\infty)}$

Ex. $x^2 + x + 1 > 0$ for all x $(a > 0\ ,\ D = -3 < 0)$

Common roots

(i) One common root of $a_1x^2 + b_1x + c_1\ ,\ a_2x^2 + b_2x + c_2 \iff$

$$\boxed{(c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1)}$$

(ii) Both roots common $\iff \dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \dfrac{c_1}{c_2}$

(iii) Rational coeffs, common root irrational / complex $\Rightarrow$ both common

(iv) Method : subtract to remove x^2, find the root, verify

Ex. $x^2 - 5x + 6\ ,\ x^2 - 7x + 10$: subtract $\to 2x - 4 = 0 \Rightarrow$ common root $\boxed{2}$

Ex. $x^2 + ax + b\ ,\ x^2 + bx + a\ (a \neq b)$: subtract $\to x = 1 \Rightarrow \boxed{a + b + 1 = 0}$

General quadratic in two variables

(i) $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ (second degree equation)

☆ (ii) Represents a **pair of straight lines** $\iff$

$$\boxed{\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0}$$

(iii) Treat as quadratic in x : D must be a perfect square in $y \Rightarrow$ factorise

(iv) Real lines need $h^2 \geq ab$; parallel lines : $h^2 = ab$

Ex. $2x^2 + 7xy + 3y^2 + 5x + 5y + \lambda$ is a pair of lines

$= (2x + y + 1)(x + 3y + 2) \Rightarrow \lambda = \boxed{2}$

Check with Δ : $2 \cdot 3 \cdot 2 + 2 \cdot \frac{5}{2} \cdot \frac{5}{2} \cdot \frac{7}{2} - 2 \cdot \frac{25}{4} - 3 \cdot \frac{25}{4} - 2 \cdot \frac{49}{4} = 0$

Coefficients & roots of n-degree equation

☆ (i) Cubic $ax^3+bx^2+cx+d=0$: $\sum\alpha=-\frac{b}{a}$, $\sum\alpha\beta=\frac{c}{a}$, $\alpha\beta\gamma=-\frac{d}{a}$

(ii) Quartic : $\sum\alpha=-\frac{b}{a}$, $\sum\alpha\beta=\frac{c}{a}$

(iii) $\sum\alpha\beta\gamma=-\frac{d}{a}$, $\alpha\beta\gamma\delta=\frac{e}{a}$

(iv) General : r at a time $=(-1)^r\frac{a_{n-r}}{a_n}$

(v) $\sum\alpha^2=(\sum\alpha)^2-2\sum\alpha\beta$; $\sum\frac{1}{\alpha}=\frac{\sum\alpha\beta}{\alpha\beta\gamma}$

(vi) $\sum\alpha^3=e_1^3-3e_1e_2+3e_3$

(vii) A.P. roots : middle $=-\frac{b}{3a}$; G.P. roots : middle$^3=-\frac{d}{a}$

Ex. x^3-2x^2+3x-4 : $\sum\alpha^2=4-6=$ $\boxed{-2}$, $\sum\frac{1}{\alpha}=$ $\boxed{\frac{3}{4}}$

Ex. $x^3-6x^2+11x-6$ (roots $1,2,3$) : $\sum\alpha^3=216-198+18=$ $\boxed{36}$

Ex. Roots of $x^3-9x^2+kx-15$ in A.P. : middle root 3

$27-81+3k-15=0\Rightarrow k=$ $\boxed{23}$

Rational roots & Descartes rule

(i) Integer coefficients, root $\frac{p}{q}$ (lowest terms) $\Rightarrow p\mid a_0$ and $q\mid a_n$

(ii) Descartes : positive roots $\le$ sign changes in $P(x)$

(iii) Negative roots $\le$ sign changes in $P(-x)$

(iv) Odd degree real polynomial has at least one real root

Ex. $2x^3-3x^2-3x+2=0$: try $x=2\Rightarrow(x-2)(2x-1)(x+1)$

roots $\boxed{2,\ \frac{1}{2},\ -1}$

Ex. $x^3-3x+1=0$: 2 sign changes in $P(x)$, 1 in $P(-x)$

$\Rightarrow$ 3 real roots $\boxed{-1.88,\ 0.35,\ 1.53}$

Graph of quadratic function : $f(x) = ax^2 + bx + c$

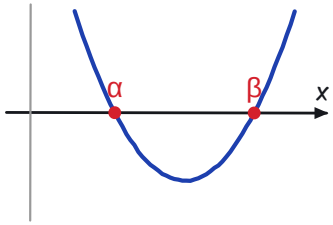
☆ (i) Graph is a parabola ; axis $x = -\frac{b}{2a}$, vertex $\left(-\frac{b}{2a}, -\frac{D}{4a}\right)$

(ii) $a > 0$: opens up , minimum at vertex

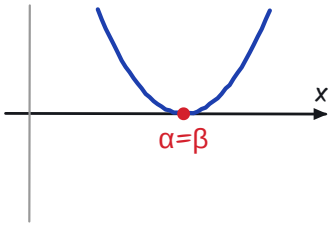
(iii) $a < 0$: opens down , maximum at vertex

(iv) $D > 0$: cuts x-axis at 2 points ; $D = 0$: touches

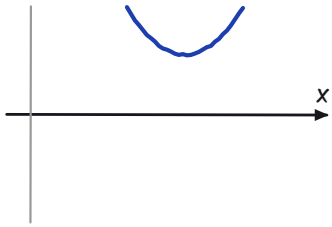
(v) $D < 0$: no real root ; y-intercept $= c$



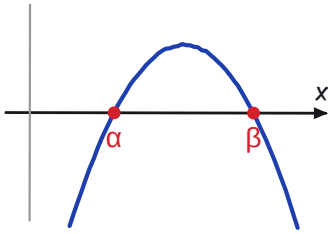
$a > 0$, $D > 0$



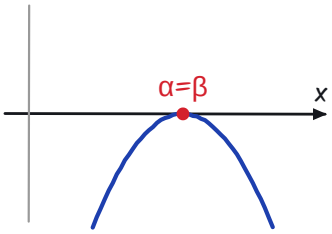
$a > 0$, $D = 0$



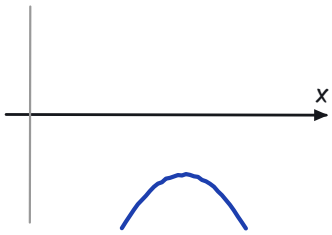
$a > 0$, $D < 0$



$a < 0$, $D > 0$



$a < 0$, $D = 0$



$a < 0$, $D < 0$

Ex. $x^2 - 6x + 13$: vertex $(3, 4)$, minimum 4 (4)

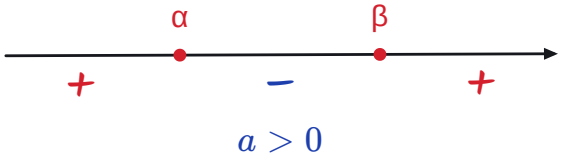
Ex. $-2x^2 + 8x + 3$: vertex $(2, 11)$, maximum (11)

Sign of quadratic expression : $f(x) = ax^2 + bx + c$

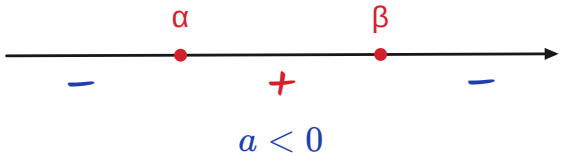
☆ (i) $D < 0$: $f(x)$ has the sign of a for all x

(ii) $D = 0$: sign of a for all $x \neq \alpha$

(iii) $D > 0$: sign of a outside the roots
opposite sign of a between roots



$a > 0$



$a < 0$

Ex. $x^2 - 5x + 6$ at $x = 2.5$: $-0.25 < 0$ (between roots $2, 3$)

Ex. $x^2 + x + 1$: $a > 0$, $D = -3 < 0 \Rightarrow$ positive for all x

Ex. $kx^2 - kx + 1 > 0$ for all x : $k = 0$ ok ; $k > 0$, $D = k^2 - 4k < 0$

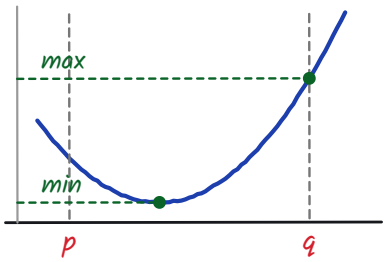
$\Rightarrow$ $0 \le k < 4$

(iv) $f(x) \ge 0$ for all $x \iff a > 0$ and $D \le 0$ (or $a = b = 0$, $c \ge 0$)

Range of quadratic function

☆ (i) $x \in \mathbb{R}$, $a > 0$: range $\left[-\frac{D}{4a}, \infty\right)$

(ii) $x \in \mathbb{R}$, $a < 0$: range $\left(-\infty, -\frac{D}{4a}\right]$



(iii) $x \in [p, q]$: vertex $x_v = -\frac{b}{2a}$ inside $\Rightarrow$ one extreme is $f(x_v)$, other is the larger (smaller) of $f(p), f(q)$

(iv) x_v outside $[p, q]$: range between $f(p)$ and $f(q)$

(v) Rational $y = \frac{f(x)}{g(x)}$: cross multiply, quadratic in x , put $D \ge 0$

Ex. $x^2 - 4x + 3$ on $[0, 5]$: vertex $x = 2$ inside , min -1 , max $\max(3, 8) = 8$

range $[-1, 8]$

Ex. $x^2 - 4x + 3$ on $[3, 5]$: increasing $\Rightarrow$ $[0, 8]$

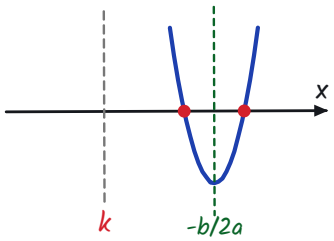
Ex. $-x^2 + 2x + 3$ on $[-1, 2]$: vertex $x = 1$, max 4 , min $\min(0, 3) = 0$

range $[0, 4]$

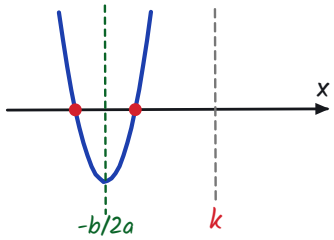
Ex. $y = \frac{x^2 + x + 1}{x^2 - x + 1}$: $(y - 1)x^2 - (y + 1)x + (y - 1) = 0$

$D \ge 0 \Rightarrow y \in$ $\left[\frac{1}{3}, 3\right]$

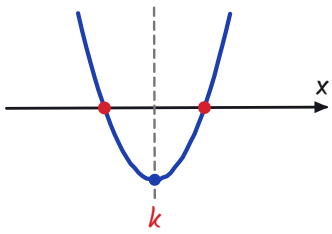
Location of roots : $f(x) = ax^2 + bx + c$, $a > 0$



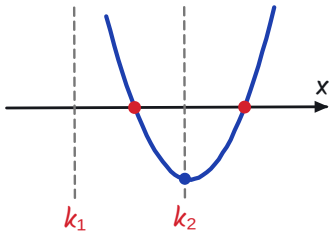
(i) $D \geq 0$, $-\frac{b}{2a} > k$, $f(k) > 0$



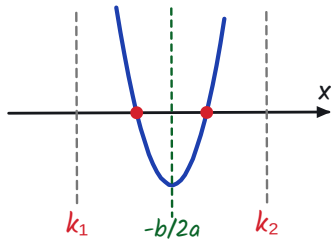
(ii) $D \geq 0$, $-\frac{b}{2a} < k$, $f(k) > 0$



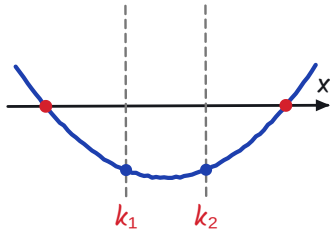
(iii) $f(k) < 0$



(iv) $f(k_1)\, f(k_2) < 0$



(v) $D \geq 0$, $k_1 < -\frac{b}{2a} < k_2$, $f(k_1), f(k_2) > 0$



(vi) $f(k_1) < 0$, $f(k_2) < 0$

(vii) Opposite signs : $c/a < 0$; both > 0 : $D \geq 0$, $S > 0$, $P > 0$

Ex. $x^2 - 2mx + m^2 - 1 = 0$, roots in $(-2, 4)$: $f(-2) > 0$, $f(4) > 0$

$(m+1)(m+3) > 0$, $(m-3)(m-5) > 0$

with $-2 < m < 4 \Rightarrow$ $-1 < m < 3$

Ex. $x^2 - 4x + k = 0$ one root in $(1, 2)$: $(k-3)(k-4) < 0 \Rightarrow$ $3 < k < 4$

Ex. 2 lies between roots of $x^2 - 2kx + 3k = 0$: $f(2) = 4 - k < 0 \Rightarrow$ $k > 4$

Equation in terms of roots of another equation

☆ (i) Method 1 : find S' , P' using S, P $\Rightarrow$ $x^2 - S'x + P' = 0$

☆ (ii) Method 2 : $y = g(\alpha)$, express $\alpha = g^{-1}(y)$, put in the given equation

(iii) Roots $\alpha + h$: $x \to x - h$; roots $k\alpha$: $x \to \frac{x}{k}$; roots $\frac{1}{\alpha}$: $x \to \frac{1}{x}$

(iv) Roots α^2, β^2 : $S' = S^2 - 2P$, $P' = P^2$

(v) Roots α^3, β^3 : $S' = S^3 - 3SP$, $P' = P^3$

(vi) Roots $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$: $S' = \frac{S^2 - 2P}{P}$, $P' = 1$

(vii) Roots $\frac{\alpha+1}{\alpha-1}$: $x = \frac{y+1}{y-1}$ in the given equation

(viii) Roots S and P themselves : $x^2 - (S+P)x + SP = 0$

Ex. $x^2 - 5x + 3 = 0$: roots $\frac{1}{\alpha}, \frac{1}{\beta} \Rightarrow$ $3x^2 - 5x + 1 = 0$

Ex. $x^2 - 5x + 3 = 0$: roots α^2, β^2 : $S' = 25 - 6 = 19$, $P' = 9$

$\Rightarrow$ $x^2 - 19x + 9 = 0$

Ex. $x^2 - 5x + 3 = 0$: roots α^3, β^3 : $S' = 80$, $P' = 27$

$\Rightarrow$ $x^2 - 80x + 27 = 0$

Ex. $x^2 - 5x + 3 = 0$: roots $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$: $S' = \frac{19}{3}$, $P' = 1$

$\Rightarrow$ $3x^2 - 19x + 3 = 0$

Ex. $x^2 - 5x + 3 = 0$: roots $\frac{\alpha+1}{\alpha-1}$: put $x = \frac{y+1}{y-1}$

$(y+1)^2 - 5(y+1)(y-1) + 3(y-1)^2 = 0 \Rightarrow$ $y^2 + 4y - 9 = 0$

Ex. $x^2 - 3x + 1 = 0$: roots $\alpha + 2, \beta + 2$: replace x by $x - 2$

$\Rightarrow$ $x^2 - 7x + 11 = 0$

Ex. $x^2 - 5x + 3 = 0$: roots S, P i.e. $5, 3 \Rightarrow$ $x^2 - 8x + 15 = 0$

Radical & fractional equations

(i) Isolate the radical, square, solve, then **verify** every root

(ii) Fractions : multiply by the LCD, exclude roots that make a denominator 0

Ex. $x = \sqrt{6+x} \Rightarrow x^2 - x - 6 = 0 \Rightarrow x = 3$ ($x = -2$ rejected) $\Rightarrow$ 3

Ex. $\sqrt{x+1} + \sqrt{x-2} = 3$: $\sqrt{x+1} = 3 - \sqrt{x-2}$, square

$6\sqrt{x-2} = 6 \Rightarrow$ $x = 3$ (verify : $2 + 1 = 3$)

Ex. $\frac{x+1}{x-1} + \frac{x-1}{x+1} = \frac{10}{3}$: $6x^2 + 6 = 10x^2 - 10 \Rightarrow x =$ ± 2

Newton sums & transformation of roots

(i) Cubic : $p_k = e_1 p_{k-1} - e_2 p_{k-2} + e_3 p_{k-3}$ $(p_k = \sum \alpha^k)$

(ii) Roots $\alpha + h$: replace x by $x - h$; roots $-\alpha$: replace x by $-x$

(iii) Roots $\frac{1}{\alpha}$: reverse the order of coefficients

(iv) Roots $k\alpha$: replace x by $\frac{x}{k}$

Ex. $x^3 - 6x^2 + 11x - 6$: $p_1 = 6$, $p_2 = 14$, $p_3 = 36$

$p_4 = 6 \cdot 36 - 11 \cdot 14 + 6 \cdot 6 =$ $\boxed{98}$

Ex. Each root increased by 1 : roots $2, 3, 4 \Rightarrow$ $\boxed{x^3 - 9x^2 + 26x - 24}$

Ex. Roots $\frac{1}{\alpha}$ of $x^3 - 6x^2 + 11x - 6$: $\boxed{6x^3 - 11x^2 + 6x - 1 = 0}$

Ex. $x^2 - 2kx + k^2 - 1 = 0$: $D = 4 > 0 \Rightarrow$ real, distinct for all k

Equations reducible to quadratic

(i) $t = f(x)$: $(x^2 + x)^2 - 2(x^2 + x) - 24 = 0 \Rightarrow t^2 - 2t - 24 = 0$

(ii) $x + \frac{1}{x} = t$: $x^2 + \frac{1}{x^2} = t^2 - 2$ (reciprocal equations)

(iii) a^x type : $4^x - 5 \cdot 2^x + 4 = 0$, $t = 2^x > 0$

(iv) $|x|$ type : $|x|^2 = x^2$, put $t = |x| \geq 0$ and reject negative t

Ex. $t = 6, -4$: $x^2 + x = 6 \Rightarrow x = 2, -3$; $x^2 + x = -4 : D < 0$

real roots $\boxed{2, -3}$

Ex. $4^x - 5 \cdot 2^x + 4 = 0$: $t = 1, 4 \Rightarrow x =$ $\boxed{0, 2}$

Ex. $x^2 - 3|x| + 2 = 0$: $|x| = 1, 2 \Rightarrow x =$ $\boxed{\pm 1, \pm 2}$

Ex. $(x^2 - x)^2 - 8(x^2 - x) + 12 = 0$: $x^2 - x = 2, 6$

$x = 2, -1, 3, -2$, sum $=$ $\boxed{2}$

Good Ques. : JEE Main style

Q1. α, β roots of $2x^2 - 7x + 3 = 0$: $\alpha^2 + \beta^2 = ?$ — Ans : $\frac{37}{4}$

Q2. $x^2 + 2(k+2)x + 9k = 0$ has equal roots : values of k ? — Ans : $1, 4$

Q3. Remainders $4, 10$ on $(x-1), (x-2)$: on $(x-1)(x-2)$? — Ans : $6x - 2$

Q4. $x^2 - 5x + k$, $x^2 - 7x + 2k$ share a root , $k \neq 0$: $k = ?$ — Ans : 6

Q5. Maximum value of $3 + 4x - 2x^2$? — Ans : 5

Q6. Sum of squares of all real roots of $x^2 - 5|x| + 6 = 0$? — Ans : 26

Q7. Integers k with exactly one root of $x^2 - 5x + k = 0$ in $(1, 2)$? — Ans : 1

Q8. $x^3 - 7x + 6 = 0$ has roots α, β, γ : $\alpha^2 + \beta^2 + \gamma^2 = ?$ — Ans : 14

Good Ques. (Advanced) : JEE Adv style

Q1. α, β roots of $x^2 - x - 1 = 0$: $\alpha^{10} + \beta^{10} = ?$ — Ans : 123

Q2. $3x^2 - 5xy - 2y^2 - x + 9y + k = 0$ is a pair of lines : $k = ?$ — Ans : -4

Q3. Sum of all real roots of $(x^2 - x)^2 - 8(x^2 - x) + 12 = 0$? — Ans : 2

Q4. Roots of $x^3 - 9x^2 + kx - 15 = 0$ are in A.P. : $k = ?$ — Ans : 23

Q5. 2 lies between the roots of $x^2 - 2kx + 3k = 0$: set of k ? — Ans : $k > 4$

Q6. Number of real roots of $x^3 - 3x + 1 = 0$? — Ans : 3

Kaunsa method lagaun ?

Roots / nature of roots	$\Rightarrow D = b^2 - 4ac$
$\alpha^2 + \beta^2$, $\alpha^3 + \beta^3$, $\frac{1}{\alpha} + \frac{1}{\beta}$	$\Rightarrow S$ aur P me badlo
New equation with roots $f(\alpha), f(\beta)$	$\Rightarrow$ naya S' , P' nikalo
Remainder on $(x-a)(x-b)$	$\Rightarrow px+q$ lo, $P(a)$, $P(b)$ lagao
Repeated root	$\Rightarrow P(\alpha) = P'(\alpha) = 0$
Common root	$\Rightarrow$ subtract karo ya $(c_1a_2 - c_2a_1)^2 = \ldots$
Pair of straight lines	$\Rightarrow \Delta = 0$
Max / min , range	$\Rightarrow$ vertex $-\frac{D}{4a}$, ya $D \geq 0$
Roots ki location	$\Rightarrow D$, $-\frac{b}{2a}$, $f(k)$ ka sign
Higher degree	$\Rightarrow$ Vieta + substitution

Last minute formula strip

$x = \frac{-b \pm \sqrt{D}}{2a}$	$S = -\frac{b}{a}$, $P = \frac{c}{a}$	$\alpha^3 + \beta^3 = S^3 - 3SP$
vertex $\left(-\frac{b}{2a}, -\frac{D}{4a}\right)$	$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$	$R = P(a)$ for $(x-a)$

Dhyan rakho : common galtiyan

(1) Squaring ke baad roots verify karo

(2) Common factor cancel mat karo, root chala jaata hai

(3) $D = 0$ means equal roots, not no roots

(4) $|x|$ me $t = |x| \geq 0$, negative t reject

(5) Location of roots me $D \geq 0$ aur vertex ki condition bhi lagao

(6) Range nikalte time x ke real hone ke liye $D \geq 0$

Hints (separately) : JEE Main

Q1 $S = \frac{7}{2}$, $P = \frac{3}{2}$, $S^2 - 2P$.

Q2 $D = 0 : 4(k+2)^2 - 36k = 0$.

Q3 Remainder $px + q : p + q = 4$, $2p + q = 10$.

Q4 Subtract the two equations $\rightarrow$ common root $x = \frac{k}{2}$, put back.

Q5 Vertex at $x = -\frac{b}{2a}$, $a < 0$.

Q6 Put $|x| = t$: roots $\pm 2, \pm 3$.

Q7 $f(1)f(2) < 0$ and k integer ; check end values separately.

Q8 $\sum \alpha = 0$, $\sum \alpha\beta = -7 : \sum \alpha^2 = (\sum \alpha)^2 - 2\sum \alpha\beta$.

Hints (Advanced) : JEE Adv

Q1 Recurrence $L_n = L_{n-1} + L_{n-2}$ with $S = 1$, $P = -1$.

Q2 Factorise as $(x - 2y + 1)(3x + y + m)$ ya $\Delta = 0$ lagao.

Q3 Put $t = x^2 - x$, solve, then solve each for x.

Q4 Middle root $= -\frac{b}{3a} = 3$, put in the equation.

Q5 $f(2) < 0$; also check $D > 0$.

Q6 Descartes rule + $P(-x)$; check sign of P at few points.

Advanced me kya dhyan rakhein

(i) Symmetric expressions ko hamesha S aur P me likho

(ii) Location of roots me end points aur vertex dono check karo

(iii) Pair of lines : $\Delta = 0$ ke saath $h^2 \geq ab$ bhi dekho

(iv) Higher degree equation me Vieta ke saath rational root test